Action | Citi Research

26 Apr 2026 21:10:44 ET | 25 pages

Taiwan Semiconductors

TPU Evolution: From Compute to Coordination - Opportunities for Mediatek, GUC and Aspeed

Laura (Chia Yi) Chen AC
+886-2-8726-9090
laura.cy.chen@citi.com

Jack Chen
+886-2-8726-9091
jack1.chen@citi.com

Nicholas Lai
+886-2-8726-9093
nicholas.lai@citi.com

Michael Hung
+886-2-8726-9092
michael.hung@citi.com

CITI'S TAKE

At Google Cloud Next '26, TPU evolution highlights a shift toward system-level efficiency rather than pure compute scaling. The latest generation splits training and inference into two architectures: v8t for high-throughput training and v8i for low-latency, high-utilization inference, particularly for agentic AI workloads. **Meanwhile, CPU resources such as Axion are increasingly critical, handling orchestration, scheduling, and dynamic routing.** Interconnect (ICI) design is also evolving — from bandwidth-driven scaling in v8t to latency-optimized communication in v8i — reflecting different workload requirements. We believe MediaTek is contributing to SerDes chiplet design for future TPU platforms, with momentum accelerating into 4Q26. In parallel, the growing importance of Axion CPUs should drive upside for GUC, as the back-end service provider. And Aspeed stands to benefit from expanding BMC demand as AI data center architectures become more complex and system-centric.

Mediatek to benefit from emerging COT business model in TPU — Mediatek has been working with Google on TPU SerDes I/O die design and turnkey solution service. Under the Customer Owned Tools (COT) model, Google owns the full TPU stack from architecture, IP, and system design — while outsourcing implementation to design partners and manufacturing to TSMC. While Mediatek has limited access to core TPU silicon, it has growing opportunity in control-plane compute, I/O integration, and edge inference chips. For the upcoming earnings call this week, we expect Mediatek to reiterate its more than US$1bn AI ASIC revenue contribution for this year and multiple-billion revenue opportunity for the next few years. With better projects visibility next year, we are now looking for US$10/14bn AI ASIC revenue in 2027/2028 and raise our target price to NT$2,800. Reiterate Buy.

GUC to benefit from the CPU renaissance — Within Google's new AI framework, the Axion CPU is playing a more important role as a system orchestrator. It manages task scheduling, data routing, and coordination across TPU clusters which is critical in future AI workloads like agent-based inference, where execution is dynamic and multi-step. Going forward, the TPU defines performance ceiling, CPU defines utilization. GUC has secured the CPU design service business and start contribution in late 2025 and the company has also secured the next generation project into next two years. With more N3 capacity support from TSMC, we expect more substantial revenue increase in 2027 and 2028. Reiterate Buy with TP to NT$5,050.

Aspeed sees structural upside on BMC — In Google's latest TPU pod architectures, the role of CPUs has clearly evolved from simple control-plane processors into system-level orchestrators that actively participate in workload scheduling, telemetry, and data movement. As a result, BMC deployment scales alongside system complexity, with each server node or TPU tray continuing to require dedicated management control to ensure fault isolation, remote debugging, and operational resilience. Aspeed already see an upbeat 1Q26 sales, we expect continuous earnings upward revision. We reiterate Buy with TP to NT$22,500.

See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

Data Summary

						Rating		Short-Term	Target Price						Current Fiscal Year EPS		Next Fiscal Year EPS	
Company	Ticker	Ccy	Price	Mkt Cap (M)	Date & Time	Old	New	View	Old	New	ESPR (%)	Div Yld (%)	ETR (%)	Last Rpt Yr	Old	New	Old	New
Aspeed Technology	5274.TWO	NT$	16,370.00	618,830	24 Apr 15:00	1	nc	-	13,250.00	22,500.00	37.4	0.5	37.9	Dec-25	158.80	189.20	264.84	284.74
GUC	3443.TW	NT$	4,035.00	540,738	24 Apr 13:30	1H	nc	Upside	2,725.00	5,050.00	25.2	0.4	25.6	Dec-25	50.68	45.78	82.72	83.20
MediaTek	2454.TW	NT$	2,435.00	3,905,507	24 Apr 13:30	1	nc	Upside	2,125.00	2,800.00	15.0	2.2	17.2	Dec-25	60.66	63.24	85.02	92.25

1 = Buy, 2 = Neutral, 3 = Sell, H = High Risk

ESPR = Expected Share Price Return, ETR = Expected Total Return, nc = no change

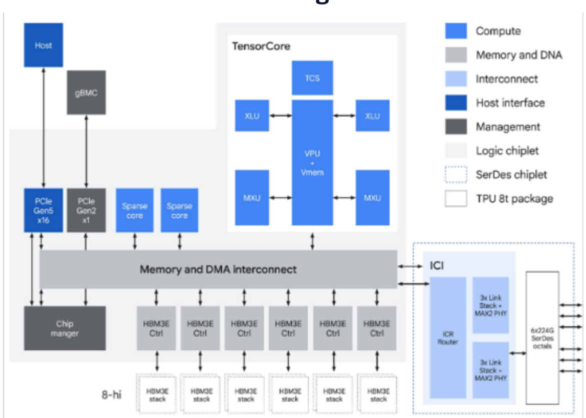
Source: Citi Research

^Catalyst Watch

TPU Evolution: From Compute to Coordination-Opportunities for Mediatek, GUC and Aspeed

Google just hosted its Cloud Next '26 last week and a world of Agentic AI. At the heart of this evolution is the Gemini Enterprise Agent Platform, which allows organizations to build "AI Agents" equipped with specialized skills. To power sophisticated agents, Google unveiled its next-generation infrastructure, including the TPU v8 family. It is worth noting that the evolution from Google TPU v7 Ironwood to Google TPU v8t and Google TPU v8i is showing **AI infrastructure has moved from a compute-centric model to a coordination-centric model**, where CPU, interconnect, and networking determine system performance more than raw FLOPs.

Figure 1. TPU v8t ASIC Block Diagram



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Google

Figure 2. TPU v8i ASIC Block Diagram

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Google

Figure 3. TPU v8t v.s. TPU v8i at a glance

Feature	TPU v8t	TPU v8i
Primary Workload	Large-scale pre-training	Sampling, serving, and reasoning
Network Topology	3D torus	Boardfly
Specialized Chip Features	SparseCore (Embeddings) & LLM Decoder Engine	CAE (Collectives Acceleration Engine)
HBM Capacity	216 GB	288 GB
On-Chip SRAM (Vmem)	128 MB	384 MB
Peak FP4 PFLOPs	12.6	10.1
HBM Bandwidth	6,528 GB/s	8,601 GB/s (~1.3x of TPU 8t)
CPU Header	Arm Axion	Arm Axion

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Google

In earlier generations like v7, the focus was on enabling large models to run by scaling memory and compute. By v8t, Google improves compute throughput and interconnect bandwidth to support efficient large-scale training. But with v8i, the emphasis changes fundamentally: **latency, orchestration, and I/O efficiency become the dominant design constraints**, reflecting the rise of agent-based, multi-step inference workloads.

Mediatek, maintaining Buy with TP to NT$2,800

Mediatek has been working with Google on AI ASIC for two years. While Mediatek has limited access to core TPU silicon, it has growing opportunity in control-plane compute, I/O integration, and edge inference chips. Some in the market believe that Mediatek is in charge of the v8t design; **however, considering the schedule and the chip functionality, we believe Mediatek is likely to be in charge of the v8i chiplet and expect the momentum to pick up in 4Q26.**

For the upcoming earnings call this week, we expect Mediatek to reiterate its more than US$1bn AI ASIC revenue contribution for this year and multiple-billion revenue opportunity for the next few years. **Considering more complex design for next generation N2 AI ASIC project with more advanced SerDes and larger die size chiplet design, for the COT business model projects, we expect various parties may provide different solutions including CPO design.** And the N2 project contribution will be more substantially in 2H28. We are now looking for US$10/14bn AI ASIC revenue in 2027/2028 for Mediatek.

Figure 4. Mediatek – Estimates Revisions

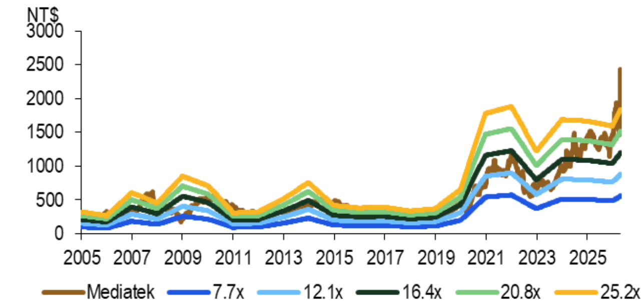
	2026E			2027E			2028E		
(NT$mn)	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.
Sales	594,690	569,770	4.4%	950,628	870,838	9.2%	1,402,154	1,129,975	24.1%
YoY growth	0%	-4%		60%	53%		47%	30%	
Gross profit	274,075	261,835	4.7%	434,207	396,841	9.4%	632,234	510,590	23.8%
Opex	177,342	169,919	4.4%	284,822	260,961	9.1%	418,914	337,747	24.0%
Operating profit	96,733	91,916	5.2%	149,386	135,880	9.9%	213,320	172,843	23.4%
Pre-tax profit	119,074	114,257	4.2%	173,275	159,769	8.5%	238,263	197,786	20.5%
Net income	100,674	96,570	4.3%	146,860	135,353	8.5%	202,232	167,745	20.6%
EPS (NT$)	63.24	60.66	4.3%	92.25	85.02	8.5%	127.03	105.37	20.6%
Gross margin	46.1%	46.0%	+0.1 ppt	45.7%	45.6%	+0.1 ppt	45.1%	45.2%	-0.1 ppt
Opex ratio	29.8%	29.8%	-0.0 ppt	30.0%	30.0%	-0.0 ppt	29.9%	29.9%	-0.0 ppt
Operating margin	16.3%	16.1%	+0.1 ppt	15.7%	15.6%	+0.1 ppt	15.2%	15.3%	-0.1 ppt
Net margin	16.9%	16.9%	-0.0 ppt	15.4%	15.5%	-0.1 ppt	14.4%	14.8%	-0.4 ppt

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Given better ASIC visibility into 2027 and more value added for the next generation project. We raise our 2026/27/28 earnings projection for Mediatek by 4%/9%/21%. In the near term, while weakness in smartphones may persist and we are looking for 7% QoQ revenue decline in 2Q and largely flat 3Q outlook, yet we expect its YoY recovery starting from late 4Q with the AI ASIC ramping up.

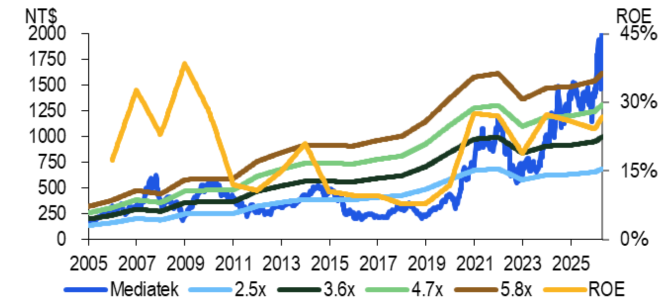
We also lift our target price to NT$2,800 from NT$2,125, based on 25x of our 2027/2028 EPS as we estimate Mediatek will transform from a consumer electronics IC/smartphone SoC maker to now cloud AI ASIC service provider. With longer AI ASIC cycle and visibility, we move our valuation base to 2027/2028 with unchanged 25x PER target. We believe the negatives on smartphone have been factored in.

Figure 5. Mediatek – Forward P/E band



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, company data

Figure 6. Mediatek – Forward P/B band & ROE



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, company data

Figure 7. Mediatek – Forecast Summary

NT$mn	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Net sales	**153,312**	**150,369**	**142,097**	**150,188**	**149,208**	**138,446**	**141,760**	**165,275**	**179,097**	**201,541**	**254,416**	**315,574**	**595,966**	**594,690**	**950,628**	**1,402,154**
Gross profit	**73,809**	**73,878**	**66,112**	**69,281**	**68,585**	**64,101**	**66,362**	**75,027**	**80,910**	**90,216**	**117,528**	**145,554**	**283,080**	**274,075**	**434,207**	**632,234**
OPEX	43,756	44,499	43,924	47,431	45,658	41,534	41,394	48,756	54,625	60,462	76,325	93,410	179,610	177,342	284,822	418,914
Operating profit	**30,053**	**29,379**	**22,188**	**21,850**	**22,928**	**22,567**	**24,968**	**26,271**	**26,285**	**29,754**	**41,203**	**52,144**	**103,470**	**96,733**	**149,386**	**213,320**
Total Non-OP	4,500	3,849	7,772	5,298	5,239	5,497	5,761	5,844	5,854	5,930	6,008	6,096	21,418	22,341	23,889	24,943
Pre-tax profit	34,553	33,228	29,960	27,147	28,167	28,064	30,728	32,115	32,140	35,684	47,211	58,240	124,888	119,074	173,275	238,263
Income tax	5,024	5,163	4,509	4,074	4,169	4,154	4,548	4,753	4,757	5,281	6,987	8,620	18,770	17,623	25,645	35,263
Net profit	**29,325**	**27,848**	**25,221**	**22,925**	**23,799**	**23,712**	**25,986**	**27,177**	**27,189**	**30,210**	**40,032**	**49,429**	**105,319**	**100,674**	**146,860**	**202,232**
EPS (NT$)	**18.43**	**17.50**	**15.84**	**14.40**	**14.95**	**14.90**	**16.32**	**17.07**	**17.08**	**18.98**	**25.15**	**31.05**	**66.17**	**63.24**	**92.25**	**127.03**
Margins (%)																
Gross profit	**48.1**	**49.1**	**46.5**	**46.1**	**46.0**	**46.3**	**46.8**	**45.4**	**45.2**	**44.8**	**46.2**	**46.1**	**47.5**	**46.1**	**45.7**	**45.1**
OPEX to Sales Ratio	28.5	29.6	30.9	31.6	30.6	30.0	29.2	29.5	30.5	30.0	30.0	29.6	30.1	29.8	30.0	29.9
Operating profit	**19.6**	**19.5**	**15.6**	**14.5**	**15.4**	**16.3**	**17.6**	**15.9**	**14.7**	**14.8**	**16.2**	**16.5**	**17.4**	**16.3**	**15.7**	**15.2**
Pre-tax profit	22.5	22.1	21.1	18.1	18.9	20.3	21.7	19.4	17.9	17.7	18.6	18.5	21.0	20.0	18.2	17.0
Net profit	**19.1**	**18.5**	**17.7**	**15.3**	**15.9**	**17.1**	**18.3**	**16.4**	**15.2**	**15.0**	**15.7**	**15.7**	**17.7**	**16.9**	**15.4**	**14.4**
Y/Y(%)																
Net sales	14.9	18.1	7.8	8.8	(2.7)	(7.9)	(0.2)	10.0	20.0	45.6	79.5	90.9	12.3	(0.2)	59.9	47.5
Gross profit	5.6	18.9	2.7	3.4	(7.1)	(13.2)	0.4	8.3	18.0	40.7	77.1	94.0	7.5	(3.2)	58.4	45.6
OPEX	16.0	19.7	8.5	4.0	4.3	(6.7)	(5.8)	2.8	19.6	45.6	84.4	91.6	11.6	(1.3)	60.6	47.1
Operating profit	(6.6)	17.7	(7.0)	2.0	(23.7)	(23.2)	12.5	20.2	14.6	31.8	65.0	98.5	1.0	(6.5)	54.4	42.8
Pre-tax profit	(3.1)	13.8	5.3	3.6	(18.5)	(15.5)	2.6	18.3	14.1	27.2	53.6	81.3	4.5	(4.7)	45.5	37.5
Net profit	(7.0)	8.3	(0.5)	(3.6)	(18.8)	(14.9)	3.0	18.5	14.2	27.4	54.1	81.9	(1.0)	(4.4)	45.9	37.7
Q/Q(%)																
Net sales	11.1	(1.9)	(5.5)	5.7	(0.7)	(7.2)	2.4	16.6	8.4	12.5	26.2	24.0				
Gross profit	10.2	0.1	(10.5)	4.8	(1.0)	(6.5)	3.5	13.1	7.8	11.5	30.3	23.8				
Operating profit	40.4	(2.2)	(24.5)	(1.5)	4.9	(1.6)	10.6	5.2	0.1	13.2	38.5	26.6				
Pre-tax profit	31.8	(3.8)	(9.8)	(9.4)	3.8	(0.4)	9.5	4.5	0.1	11.0	32.3	23.4				
Net profit	23.3	(5.0)	(9.4)	(9.1)	3.8	(0.4)	9.6	4.6	0.0	11.1	32.5	23.5				

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, company data

CPU from Host Processor to System Orchestrator, GUC and Aspeed to benefit

In Google's latest TPU pod architectures, such as TPU v8i and TPU v8t, the role of CPUs has clearly evolved from simple control-plane processors into system-level orchestrators that actively participate in workload scheduling, telemetry, and data movement. This architectural shift increases the number of CPU-associated nodes and management domains within each pod, which in turn reinforces the need for distributed, node-level out-of-band (OOB) management.

Particularly for Google Axion CPU, its backend service is done by GUC and designed by Google internally. It handles scheduling, orchestration, and coordination across compute units and memory domains, including sequencing and resource allocation; CPU would also enable low-latency response in dynamic workloads.

Given the importance of the CPU in Google's TPU architecture, we also see BMC deployment scaling alongside system complexity, with each server node or TPU tray continuing to require dedicated management control to ensure fault isolation, remote debugging, and operational resilience.

GUC – maintaining Buy with TP to NT$5,050

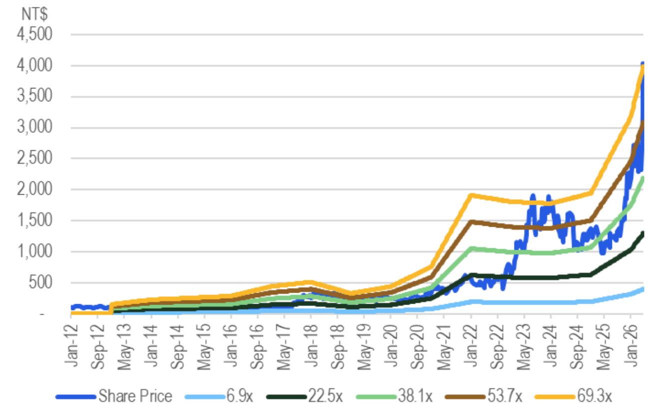
We estimate GUC's revenue from CPUs including Google and Microsoft contributes 30%+ of its 2026 revenue, and see significant upside in 2027. GUC is also working with a US EV maker on its most advanced N3 chip for both autonomous driving and robotic, which we think could fuel its growth starting from 2H27. While we lower our 2026 earnings estimates given slower crypto currency order and potential N3 wafer shortage and lower margin on less NRE revenue, we raise our 2027 estimates on stronger outlook backed by promising CPUs growth and new project from US EV clients. We also introduce our 2028 estimates. Anticipating promising growth outlook into next two years, we reiterate our Buy with TP to NT$5,050 from NT$2,725. We lift our target multiple from previous 41x to now 47x, slightly above its 3-year average PER and move our valuation base to 2027/28 average EPS, considering its better product cycle visibility and secular earnings growth trend.

Figure 8. GUC – Estimates Revisions

	2026E			2027E			2028E		
(NT$mn)	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.
Sales	56,865	58,934	-4%	94,189	88,044	7%	157,407	na	na
YoY growth	67%	82%		66%	49%		67%	na	na
Gross profit	12,652	13,637	-7%	20,546	20,034	3%	33,305	na	na
Opex	5,725	5,925	-3%	7,728	7,271	6%	12,931	na	na
Operating profit	6,927	7,712	-10%	12,818	12,764	0%	20,374	na	na
Pre-tax profit	7,207	7,992	-10%	13,098	13,044	0%	20,654	na	na
Net income	6,165	6,824	-10%	11,203	11,138	1%	17,666	na	na
EPS (NT$)	45.78	50.68	-10%	83.20	82.72	1%	131.20	na	na
Gross margin	22.2%	23.1%	-0.9 ppt	21.8%	22.8%	-0.9 ppt	21.2%	na	na
Opex ratio	10.1%	10.1%	+0.0 ppt	8.2%	8.3%	-0.1 ppt	8.2%	na	na
Operating margin	12.2%	13.1%	-0.9 ppt	13.6%	14.5%	-0.9 ppt	12.9%	na	na
Net margin	10.8%	11.6%	-0.7 ppt	11.9%	12.7%	-0.8 ppt	11.2%	na	na

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

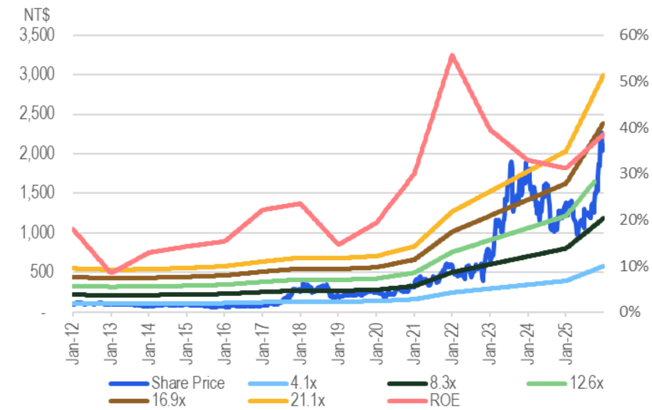
Figure 9. GUC – Forward P/E band



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 10. GUC – Forward P/B band & ROE



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 11. GUC – Forecast Summary

GUC	2025				2026				2027							
(NT$ in Mn, year-end Dec)	1Q	2Q	3Q	4Q	1QE	2QE	3QE	4QE	1QE	2QE	3QE	4QE	2025	2026E	2027E	2028E
Revenue	7,024	6,105	8,613	12,400	11,411	13,948	14,877	16,629	17,379	20,521	26,414	29,875	34,141	56,865	94,189	157,407
COGS	-5,051	-4,071	-6,527	-10,038	-8,930	-10,817	-11,549	-12,917	-13,516	-16,028	-20,694	-23,406	-25,687	-44,213	-73,643	-124,101
Depreciation costs	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-23	-22	-22	-22
Gross Profit	1,973	2,034	2,086	2,361	2,481	3,131	3,328	3,712	3,863	4,493	5,721	6,469	8,454	12,652	20,546	33,305
Operating Expense	-878	-1,090	-1,031	-1,105	-1,358	-1,437	-1,384	-1,547	-1,564	-1,724	-2,140	-2,300	-4,104	-5,725	-7,728	-12,931
SG&A expenses	-224	-253	-218	-224	-308	-321	-342	-382	-348	-349	-449	-508	-920	-1,354	-1,653	-2,765
R&D expenses	-800	-837	-813	-880	-1,050	-1,116	-1,041	-1,164	-1,217	-1,375	-1,691	-1,792	-3,330	-4,371	-6,074	-10,167
EBIT	1,095	944	1,055	1,257	1,123	1,694	1,944	2,165	2,299	2,769	3,581	4,169	4,350	6,927	12,818	20,374
Net Interest Income	45	44	43	45	18	18	18	18	18	18	18	18	177	72	72	72
Net Other Income	3	-59	-82	54	51	52	52	53	51	52	52	53	-84	208	208	208
Pre-Tax Profit	1,143	929	1,015	1,356	1,192	1,764	2,014	2,236	2,368	2,839	3,651	4,240	4,443	7,207	13,098	20,654
Tax	-181	-147	-148	-196	-172	-255	-291	-324	-343	-411	-528	-613	-673	-1,043	-1,895	-2,988
Net Profit	961	782	867	1,160	1,020	1,509	1,723	1,913	2,025	2,428	3,123	3,626	3,770	6,165	11,203	17,666
EPS (NT$)	7.14	5.81	6.44	8.61	7.57	11.21	12.80	14.21	15.04	18.04	23.19	26.93	28.01	45.78	83.20	131.20
Margins (%)																
Gross Margin	28.1%	33.3%	24.2%	19.0%	21.7%	22.4%	22.4%	22.3%	22.2%	21.9%	21.7%	21.7%	24.8%	22.2%	21.8%	21.2%
Operating Margin	15.6%	15.5%	12.2%	10.1%	9.8%	12.1%	13.1%	13.0%	13.2%	13.5%	13.6%	14.0%	12.7%	12.2%	13.6%	12.9%
Net Margin	13.7%	12.8%	10.1%	9.4%	8.9%	10.8%	11.6%	11.5%	11.7%	11.8%	11.8%	12.1%	11.0%	10.8%	11.9%	11.2%
Sequential Growth (%)																
Revenue	17%	-13%	41%	44%	-8%	22%	7%	12%	5%	18%	29%	13%	36%	67%	66%	67%
Gross Profit	-1%	3%	3%	13%	5%	26%	6%	12%	4%	16%	27%	13%	4%	50%	62%	62%
EBIT	25%	-14%	12%	19%	-11%	51%	15%	11%	6%	20%	29%	16%	14%	59%	85%	59%
Net Profit	14%	-19%	11%	34%	-12%	48%	14%	11%	6%	20%	29%	16%	9%	64%	82%	58%
EPS	14%	-19%	11%	34%	-12%	48%	14%	11%	6%	20%	29%	16%	9%	63%	82%	58%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Aspeed- maintaining Buy with TP to NT$22,500

Given more important architecture of CPU-associated nodes and management domains within each pod, which in turn reinforces the need for distributed, node-level out-of-band (OOB) management. There is increasing BMC deployment scales alongside system complexity, with each server node or TPU tray continuing to require dedicated management control to ensure fault isolation, remote debugging, and operational resilience.

In addition, the rapid increase in rack-level power density (often exceeding 100kW) and the transition to liquid cooling architectures significantly elevate system management requirements. In such environments, BMCs remain the only reliable out-of-band control interface, and their role becomes even more critical for monitoring thermals, managing failures, and ensuring system stability, often with added redundancy at the rack or system level.

We see long-term BMC demand firmly supported by increasing system scale and complexity. The evolution of CPU roles within TPU pods reinforces their necessity within a more distributed and mission-critical management framework. We see a steady expansion of BMC total addressable market (TAM), driven by hyperscale deployment trends.

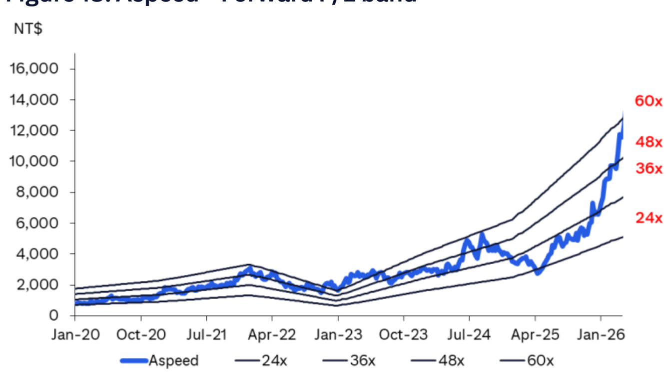
Supported by strong growth trajectory of key platforms from hyperscalers such as NVIDIA and Google, we see solid unit shipment and ASP upside for Aspeed. The transition to higher-value platforms (ie VR200, TPU V8), alongside the introduction of new BMC generations (ie AST2700), is allowing Aspeed to capture disproportionate profit expansion relative to shipment growth. We lift our 2026/2027 earnings projection and introduce our 2028 estimates by assuming over 40% YoY BMC shipment annually with ASP upside. We reiterate our Buy with higher target price of NT$22,500 from NT$13,250 based on 50x of our 2028 EPS projection, in line with its historical PER range but moving to 2028 valuation base year on clear structural growth trend.

Figure 12. Aspeed – Estimates Revisions

	2026E			2027E			2028E		
(NT$mn)	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.
Sales	16,055	14,100	13.9%	24,185	23,288	3.8%	38,453	na	na
Gross profit	11,012	9,576	15.0%	16,645	15,894	4.7%	26,184	na	na
Opex	2,311	2,278	1.5%	3,503	3,656	-4.2%	5,468	na	na
Operating profit	8,701	7,299	19.2%	13,142	12,238	7.4%	20,716	na	na
Pre-tax profit	8,940	7,503	19.1%	13,454	12,513	7.5%	21,201	na	na
Net income	7,152	6,003	19.1%	10,763	10,011	7.5%	16,961	na	na
EPS (NT$)	189.20	158.80	19.1%	284.74	264.84	7.5%	448.71	na	na
Gross margin	68.59	67.92	0.67	68.82	68.25	0.57	68.09	na	na
Opex ratio	14.40	16.15	(1.76)	14.48	15.70	(1.22)	14.22	na	na
Operating margin	54.20	51.76	2.43	54.34	52.55	1.79	53.87	na	na
Net margin	44.55	42.57	1.97	44.50	42.99	1.52	44.11	na	na

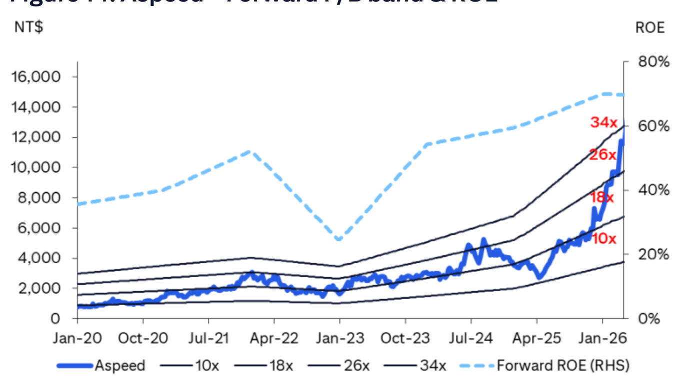
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 13. Aspeed – Forward P/E band



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, company data

Figure 14. Aspeed – Forward P/B band & ROE



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, company data

Figure 15. Aspeed – Forecast Summary

NT$mn	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	2025	2026E	2027E	2028E
Net sales	2,065	2,247	2,330	2,443	3,147	3,670	4,340	4,898	9,085	16,055	24,185	38,453
Gross profit	**1,368**	**1,525**	**1,614**	**1,672**	**2,125**	**2,479**	**3,008**	**3,400**	**6,179**	**11,012**	**16,645**	**26,184**
OPEX	341	341	382	455	516	551	595	650	1,518	2,311	3,503	5,468
Operating profit	**1,027**	**1,184**	**1,231**	**1,218**	**1,609**	**1,929**	**2,414**	**2,750**	**4,660**	**8,701**	**13,142**	**20,716**
Total Non-OP	78	(397)	275	239	57	58	62	62	195	238	312	485
Pre-tax profit	1,105	787	1,506	1,457	1,665	1,987	2,476	2,812	4,855	8,940	13,454	21,201
Income tax	218	158	292	259	333	397	495	562	928	1,788	2,691	4,240
Net profit	**887**	**629**	**1,214**	**1,198**	**1,332**	**1,589**	**1,981**	**2,250**	**3,928**	**7,152**	**10,763**	**16,961**
EPS (NT$)	**23.5**	**16.6**	**32.1**	**31.7**	**35.2**	**42.0**	**52.4**	**59.5**	**103.9**	**189.2**	**284.7**	**448.7**
Product mix (%)												
BMC	95%	95%	94%	93%	94%	94%	94%	94%	95%	94%	93%	90%
PC AV & Cupola 360	4%	4%	5%	5%	4%	4%	5%	4%	4%	4%	4%	3%
PFR	1%	1%	1%	1%	1%	1%	2%	2%	1%	1%	3%	7%
I/O Expander	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Margins (%)												
Gross profit	**66.2**	**67.9**	**69.3**	**68.5**	**67.5**	**67.5**	**69.3**	**69.4**	**68.0**	**68.6**	**68.8**	**68.1**
OPEX to Sales Ratio	16.5	15.2	16.4	18.6	16.4	15.0	13.7	13.3	16.7	14.4	14.5	14.2
Operating profit	**49.7**	**52.7**	**52.9**	**49.8**	**51.1**	**52.5**	**55.6**	**56.1**	**51.3**	**54.2**	**54.3**	**53.9**
Pre-tax profit	53.5	35.0	64.6	59.6	52.9	54.1	57.0	57.4	53.4	55.7	55.6	55.1
Net profit	**42.9**	**28.0**	**52.1**	**49.0**	**42.3**	**43.3**	**45.6**	**45.9**	**43.2**	**44.5**	**44.5**	**44.1**
Y/Y(%)												
Net sales	104	66	17	16	52	63	86	100	41	77	51	59
Gross profit	113	77	26	21	55	63	86	103	49	78	51	57
OPEX	34	22	17	21	51	62	56	43	23	52	52	56
Operating profit	166	104	29	22	57	63	96	126	60	87	51	58
Pre-tax profit	136	24	65	26	51	153	64	93	53	84	50	58
Net profit	127	24	66	27	50	153	63	88	53	82	50	58
Q/Q(%)												
Net sales	(2)	9	4	5	29	17	18	13				
Gross profit	(1)	11	6	4	27	17	21	13				
Operating profit	3	15	4	(1)	32	20	25	14				
Pre-tax profit	(4)	(29)	91	(3)	14	19	25	14				
Net profit	(6)	(29)	93	(1)	11	19	25	14				

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Adding Upside 90-Day Short-Term View on GUC (3443.TW)

Direction:	**Upside**
Duration:	Within 90 Days
Category:	Thematic driven

We are seeing strong growth momentum from GUC's CPU project with leading CSP client in the recent quarters to provide upside to its share price.

Bull/Bear: Aspeed Technology (5274.TWO)

NT$27,000.00	NT$22,500.00	NT$12,000.00
▲65% Upside	▲37% Upside	▼27% Downside

Spread 92pp
Current Price and expected returns (upside/downside) as of 24 Apr 2026

BULL Assumptions

- Higher target multiple at 60x

BASE Assumptions

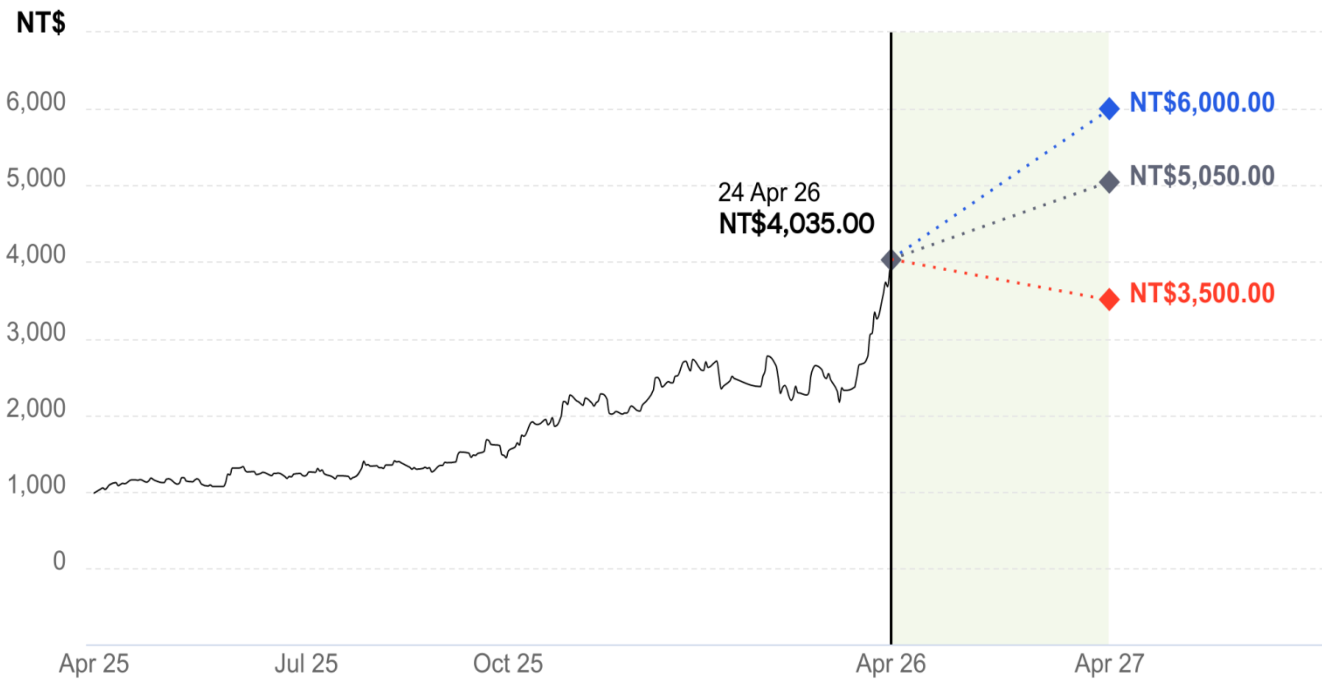
- Sales to grow by 51-77% y-o-y in 2026E-27E with OPM at 54.2-54.3%
- Target multiple at 50x

BEAR Assumptions

- Sales to grow by 20% y-o-y in 2026E-27E with OPM at 45.0%
- Target multiple at 30x

Bull/Bear: GUC (3443.TW)

NT$ 6,000.00 ▲49% Upside	NT$ 5,050.00 ▲25% Upside	NT$ 3,500.00 ▼13% Downside



Spread 62pp
Current Price and expected returns (upside/downside) as of 24 Apr 2026

BULL Assumptions

- Stronger projects demand from US CSP customers
- Stronger-than-expected NRE demand from HPC

BASE Assumptions

- Smooth project progress with US customers in advanced node demand

BEAR Assumptions

- Slower projects demand from US CSP customers
- Weaker-than-expected NRE demand from HPC

Bull/Bear: Mediatek (2454.TW)

NT$ 3,000.00 ▲23% Upside	NT$ 2,800.00 ▲15% Upside	NT$ 2,000.00 ▼18% Downside

Spread 41pp
Current Price and expected returns (upside/downside) as of 24 Apr 2026

BULL Assumptions

- Higher than expected AI ASIC development
- Stronger than expected margins expansion
- Bull case TP is based on its peak cycle valuation of 27x PER to 2027/28E EPS

BASE Assumptions

- 2027/28E GM of >45%
- 2027/28E OPM of >15%

BEAR Assumptions

- Lower than AI ASIC development
- Worse than expected margins declines
- Bear case TP is based on 18x PER (the mid-point of its historical valuation) to 2027/28E EPS

Aspeed Technology

Company description

Established in 2004 and listed in 2013, Aspeed is a leading IC design company based in Taiwan focusing on baseboard management controller (BMC) solutions. Aspeed started the business by targeting datacenter servers/white-box markets and has been working closely with server ODMs. Following the acquisition of Emulex Pilot from AVGO/Broadcom in 2016, Aspeed successfully expanded its product lines, IP pool as well as customer base and became the largest BMC supplier with c.70% of global market share. Other than BMC, Aspeed is also broadening its product portfolio by including bridge IC (BIC; or mini-BMC), platform firmware resilience (PFR) solutions and input/output (I/O) expander, aiming to maximize the dollar content it could generate per server motherboard.

Investment strategy

We rate Aspeed shares a Buy. As a leading BMC supplier with c.70%+ global market share, it has been riding the strong momentum underpinned by a recovery of general-purpose servers and upticks in AI servers. We expect the growth to continue. Apart from being in a dominant position for the supply of upcoming GB200 server BMCs with higher dollar content and better capex efficiency, new product launches including AST2700, PFR, I/O expander and Cupola 360, as well as steady growth in general-purpose servers should also act as key drivers for the company.

Valuation

We value Aspeed at NT$22,500 based on a historical mid-cycle multiple (since 2020) of 50x 2028E EPS, as we believe it fairly reflects the company’s earnings performance and industry dynamics, secular growth trends, and Aspeed’s strong supply chain position, as well as robust earnings growth amid margin expansions.

Risks

Our quant rating system rates the Aspeed stock High Risk based on its volatility over the past year, which we think is unwarranted given the company's dominance and outlook in the BMC sector. Key downside risks to our earnings forecasts and target price include: 1) weaker-than-expected growth in servers; 2) slower-than-expected CPU platform migration; 3) peer competition; and 4) limited progress in product diversification.

GUC

Company description

Founded in 1998, Global Unichip (GUC) specializes in providing IC design services, particularly in ASIC service. As a subsidiary of TSMC, which holds 34.8% of GUC, it offers a wide range of design services including spec-in and SoC integration, physical implementation, 2.5D/3D packaging technologies, mass production services, and advanced technologies such as high bandwidth memory (HBM) and chip to chip interconnect IP. With its diversified global presence – Taiwan, North America, China, Europe, South Korea, and Japan – the company offers services including mass production, NRE, customer wafer verification projects, and IP licensing. Mass production services provide customers with complete service, from back-end design completion to wafer manufacturing, testing and management.

Investment strategy

We rate GUC stock as Buy/High Risk. We are seeing strong growth from its CPU projects with two leading CSP customers. We also see strong order potential from its EV clients. Long-term, we still believe back-end design capability and 3DIC system integration will be critical to AI accelerators, which would give better opportunities for TSMC 3DFabric partners with GUC also being a beneficiary, in our view.

Valuation

Our target price for GUC of NT$5,050 is based on 47x PER to the average of our 2027/28E EPS, above its historical 3-year PER average which we believe is justified by its strong earnings outlook supported by solid demand from its AI projects with two US leading CSPs and growth upside from its US EV clients. Our target price equates to 2026/27E PB of 22x/14x.

Risks

Our quant risk rating system rates the GUC stock High Risk for its share-price volatility. Downside risks that could mean the stock trades below our target price include: 1) slower projects demand from US CSP customers, 2) lower-than-expected NRE demand from less AI ASIC projects, and 3) delayed or less turnkey revenues contribution from key customers.

Mediatek

Company description

MediaTek (MTK) is a fabless IC company that designs SoC for smartphone, smart TV, and digital media devices. It also develops chips for connectivity, networking, smart home gadgets, and automotive sectors. It was founded in 1997, is listed on the Taiwan Stock Exchange, and is headquartered in Taiwan. It has sales and research subsidiaries in mainland China, Singapore, India, the US, Japan, Korea, Denmark, and England. It primarily sells its products in China and other emerging-market countries.

Investment strategy

We rate MediaTek shares as Buy. We expect Mediatek to ride on 5G migration in emerging markets and market share gain of its high-end Dimensity series in China market as well. In particular, we are seeing strong growth potential from its AI ASIC, NTN-satellite business in the long term. We believe solid fundamentals and strong operating leverage will support Mediatek's margins outlook.

Valuation

Our NT$2,800 target price for MediaTek is based on 25x P/E to 2027/2028 EPS, which is the high end of its five-year average forward P/E given the promising outlook from its AI ASIC business starting from 2H26. AI ASIC development with CSPs should provide more upside from 2027 onwards. We also see Mediatek well prepared for 6G and 6G+NTN Satellite communication. Despite lukewarm smartphone overall deamand, we expect Mediatek to benefit from 5G migration in emerging markets and its Dimensity series should also gain shares in flagship models. Non-smartphone business is also seeing stabilizing TV and IoT demand with Mediatek outgrowing the overall market on its better technology in WiFi 7 and connectivity.

Risks

Key downside risks that could prevent the shares from reaching our target price include: 1) weaker-than-expected growth from AI ASIC business; 2) more severe competition in the smartphone segment; 3) slower overall demand related to macroeconomic risks, especially for smartphone demand; and 4) further price hikes from foundries.

If you are visually impaired and would like to speak to a Citi representative regarding the details of the graphics in this document, please call USA 1-888-500-5008 (TTY: 711), from outside the US +1-210-677-3788

Appendix A-1

ANALYST CERTIFICATION

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

GUC (3443.TW)
Ratings and Target Price History
Fundamental Research

Analyst: Laura (Chia Yi) Chen

#	Date	Rating	Target Price	Closing Price
1	19-May-23 11:01:33	*2H	*1,280.00	1,280.00
2	28-Jul-23 15:06:57	*3H	*1,540.00	1,815.00
3	10-Dec-23 22:11:06	3H	*1,440.00	1,895.00
4	25-Apr-24 19:13:09	*1H	*1,500.00	1,200.00
5	20-Jun-24 04:12:46	1H	*2,040.00	1,625.00
6	26-Jul-24 10:14:16	1H	*1,760.00	1,335.00
7	07-Feb-25 13:35:43	1H	*1,750.00	1,380.00
8	24-Apr-25 18:56:50	1H	*1,200.00	964.00
9	02-Jul-25 10:54:37	*3H	*1,225.00	1,325.00
10	07-Oct-25 23:46:05	*1H	*1,850.00	1,385.00
11	02-Jan-26 11:31:34	1H	*2,725.00	2,185.00

*Indicates Change

Rating/target price changes above reflect Eastern Time

Aspeed Technology (5274.TWO)
Ratings and Target Price History
Fundamental Research

Analyst: Laura (Chia Yi) Chen

#	Date	Rating	Target Price	Closing Price
1	06-Dec-24 08:12:52	*1	*6,000.00	4,080.00
2	05-May-25 09:17:14	1	*4,800.00	3,230.00
3	11-Jun-25 05:58:45	1	*5,600.00	4,660.00
4	05-Aug-25 09:52:42	1	*5,875.00	4,800.00
5	05-Oct-25 14:28:21	1	*6,750.00	5,460.00
6	24-Nov-25 11:34:17	1	*7,375.00	6,405.00
7	05-Jan-26 03:43:52	1	*8,235.00	7,105.00
8	05-Mar-26 09:02:05	1	*13,250.00	9,520.00

*Indicates Change

Rating/target price changes above reflect Eastern Time

Mediatek (2454.TW)

Ratings and Target Price History
Fundamental Research

Analyst: Laura (Chia Yi) Chen

	Date	Rating	Target Price	Closing Price
1	20-Apr-23 10:55:30	2	*620.00	673.00
2	28-Jul-23 11:22:19	2	*660.00	658.00
3	20-Sep-23 14:57:20	2	*735.00	759.00
4	24-Oct-23 12:02:27	*1	*935.00	816.00
5	06-Nov-23 16:50:42	1	*1,030.00	882.00
6	10-Mar-24 16:15:26	1	*1,420.00	1,230.00
7	11-Sep-24 10:08:27	1	*1,250.00	1,110.00
8	30-Oct-24 10:35:19	1	*1,420.00	1,290.00
9	02-Jan-25 08:42:43	1	*1,580.00	1,350.00
10	07-Feb-25 11:57:14	1	*1,800.00	1,525.00
11	30-Apr-25 10:15:39	1	*1,600.00	1,350.00
12	02-Jul-25 10:54:37	1	*1,450.00	1,295.00
13	30-Jul-25 08:56:32	1	*1,540.00	1,380.00
14	28-Jan-26 07:35:51	1	*2,200.00	1,780.00
15	04-Feb-26 07:24:16	1	*2,350.00	1,800.00
16	01-Apr-26 13:20:42	1	*2,125.00	1,465.00

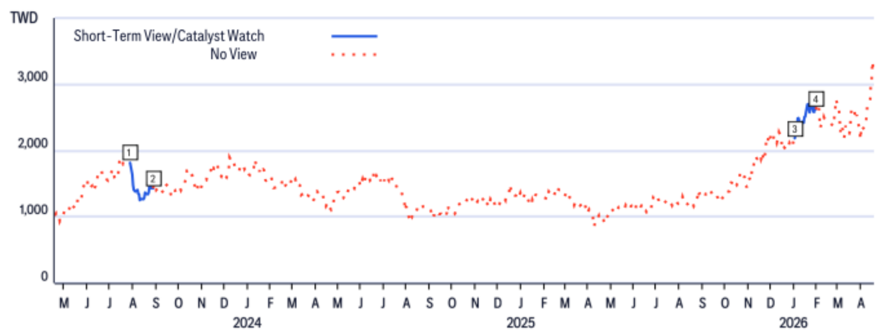
*Indicates Change

Rating/target price changes above reflect Eastern Time

GUC (3443.TW)

Short-Term View/Catalyst Watch Research

Analyst: Laura (Chia Yi) Chen



	Date	Action	Expected Direction	Duration	Closing Price
1	29-Jul-23 01:08:51	Add CW	Downside	30 Days	1,815.00
2	28-Aug-23 11:44:27	Remove CW	Downside	30 Days	1,430.00
3	02-Jan-26 06:31:34	Add STV	Upside	30 Days	2,185.00
4	01-Feb-26 19:57:44	Remove STV	Upside	30 Days	2,620.00

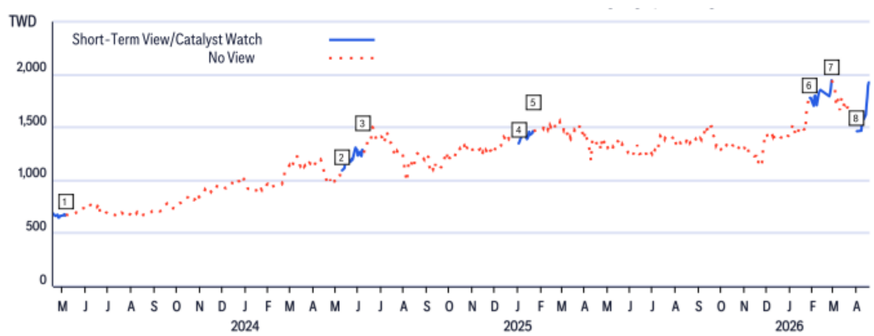
CW - Catalyst Watch , STV - Short-Term View

Rating/target price changes above reflect Eastern Time

Mediatek (2454.TW)

Short-Term View/Catalyst Watch Research

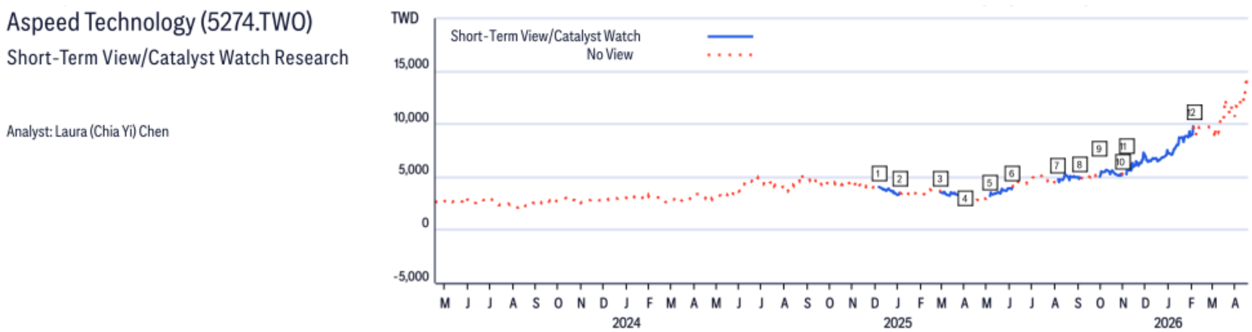
Analyst: Laura (Chia Yi) Chen



	Date	Action	Expected Direction	Duration	Closing Price
1	05-May-23 11:40:10	Remove CW	Downside	90 Days	675.00
2	10-May-24 03:08:23	Add CW	Upside	30 Days	1,095.00
3	07-Jun-24 12:03:57	Remove CW	Upside	30 Days	1,275.00
4	02-Jan-25 03:42:43	Add CW	Upside	30 Days	1,350.00
5	31-Jan-25 12:14:36	Remove CW	Upside	30 Days	1,465.00
6	28-Jan-26 02:35:51	Add STV	Upside	30 Days	1,780.00
7	27-Feb-26 12:27:51	Remove STV	Upside	30 Days	1,945.00
8	01-Apr-26 09:20:42	Add STV	Upside	90 Days	1,465.00

CW - Catalyst Watch , STV - Short-Term View

Rating/target price changes above reflect Eastern Time



	Date	Action	Expected Direction	Duration	Closing Price
1	06-Dec-24 03:12:52	Add STV	Upside	30 Days	4,080.00
2	05-Jan-25 20:28:12	Remove STV	Upside	30 Days	3,530.00
3	02-Mar-25 11:36:08	Add STV	Upside	30 Days	3,590.00
4	01-Apr-25 22:14:16	Remove STV	Upside	30 Days	3,075.00
5	05-May-25 05:17:14	Add STV	Upside	30 Days	3,230.00
6	04-Jun-25 22:24:32	Remove STV	Upside	30 Days	4,025.00
7	05-Aug-25 05:52:42	Add STV	Upside	30 Days	4,800.00
8	04-Sep-25 22:02:09	Remove STV	Upside	30 Days	4,935.00
9	30-Sep-25 12:56:21	Add STV	Upside	30 Days	5,050.00
10	30-Oct-25 22:18:17	Remove STV	Upside	30 Days	5,210.00
11	05-Nov-25 04:18:31	Add STV	Upside	90 Days	5,285.00
12	03-Feb-26 20:27:44	Remove STV	Upside	90 Days	9,790.00

CW - Catalyst Watch , STV - Short-Term View

Rating/target price changes above reflect Eastern Time

Citigroup Global Markets Inc. or its affiliates received compensation for products and services other than investment banking services from Aspeed Technology,GUC,Mediatek in the past 12 months.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, securities-related: GUC,Mediatek.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, non-securities-related: Aspeed Technology,GUC,Mediatek.

Analysts' compensation is determined by Citi Research management and Citigroup's senior management and is based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates (the "Firm"). Compensation is not linked to specific transactions or recommendations. Like all Firm employees, analysts receive compensation that is impacted by overall Firm profitability which includes investment banking, sales and trading, and principal trading revenues. One factor in equity research analyst compensation is arranging corporate access events between institutional clients and the management teams of covered companies. Typically, company management is more likely to participate when the analyst has a positive view of the company.

For financial instruments recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in such financial instruments (and any underlying instruments) and may act as principal in connection with transactions in such instruments. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Unless stated otherwise neither the Research Analyst nor any member of their team has viewed the material operations of the Companies for which an investment view has been provided within the past 12 months.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 6th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Pursuant to the Market Abuse Regulation a history of all Citi Research recommendations published during the preceding 12-month period can be accessed via Citi Velocity (https://www.citivelocity.com/cv2) or your standard distribution portal. Historical disclosures (for up to the past three years) will be provided upon request.

Citi Research Equity Ratings Distribution

	12 Month Rating			Catalyst Watch		
Data current as of 01 Apr 2026	**Buy**	**Hold**	**Sell**	**Buy**	**Hold**	**Sell**
Citi Research Global Fundamental Coverage (Neutral=Hold)	61%	32%	8%	37%	47%	16%
% of companies in each rating category that are investment banking clients	38%	41%	28%	42%	37%	36%

Guide to Citi Research Fundamental Research Investment Ratings:

Citi Research stock recommendations include an investment rating and an optional risk rating to highlight high risk stocks.

Risk rating takes into account both price volatility and fundamental criteria. Stocks will either have no risk rating or a High risk rating assigned.

Investment Ratings: Citi Research investment ratings are Buy, Neutral and Sell. Our ratings are a function of analyst expectations of expected total return ("ETR") and risk. ETR is the sum of the forecast price appreciation (or depreciation) plus the dividend yield for a stock within the next 12 months. The target price is based on a 12 month time horizon. The Investment rating definitions are: Buy (1) ETR of 15% or more or 25% or more for High risk stocks; and Sell (3) for negative ETR. Any covered stock not assigned a Buy or a Sell is a Neutral (2). For stocks rated Neutral (2), if an analyst believes that there are insufficient valuation drivers and/or investment catalysts to derive a positive or negative investment view, they may elect with the approval of Citi Research management not to assign a target price and, thus, not derive an ETR. Citi Research may suspend its rating and target price and assign "Rating Suspended" status for regulatory and/or internal policy reasons. Citi Research may also suspend its rating and target price and assign "Under Review" status for other exceptional circumstances (e.g. lack of information critical to the analyst's thesis, trading suspension) affecting the company and/or trading in the company's securities. In both such situations, the rating and target price will show as "—" and "-" respectively in the rating history price chart. Prior to 11 April 2022 Citi Research assigned "Under Review" status to both situations and prior to 11 Nov 2020 only in exceptional circumstances. As soon as practically possible, the analyst will publish a note re-establishing a rating and investment thesis. Investment ratings are determined by the ranges described above at the time of initiation of coverage, a change in investment and/or risk rating, or a change in target price (subject to limited management discretion). At times, the expected total returns may fall outside of these ranges because of market price movements and/or other short-term volatility or trading patterns. Such interim deviations will be permitted but will become subject to review by Research Management. Your decision to buy or sell a security should be based upon your personal investment objectives and should be made only after evaluating the stock's expected performance and risk.

Catalyst Watch/Short Term Views ("STV") Ratings Disclosure:
Catalyst Watch and STV Upside/Downside calls: Citi Research may also include a Catalyst Watch or STV Upside or Downside call to indicate the analyst expects the share price to rise (fall) in absolute terms over a specified period of 30 or 90 days in reaction to one or more specific near-term catalysts or events impacting the company or the market. A Catalyst Watch will be published when Analyst confidence is high that an impact to share price will occur; it will be a STV when confidence level is moderate. A Catalyst Watch or STV Upside/Downside call will automatically expire at the end of the specified 30/90 day period. The Catalyst Watch will also be automatically removed if share price performance (calculated at market close) exceeds 15% against the direction of the call (unless over-ridden by the analyst). The analyst may also remove a Catalyst Watch or STV call prior to the end of the specified period in a published research note. A Catalyst Watch/STV Upside or Downside call may be different from and does not affect a stock's fundamental equity rating, which reflects a longer-term total absolute return expectation. For purposes of FINRA ratings-distribution-disclosure rules, a Catalyst Watch/STV Upside call corresponds to a buy recommendation and a Catalyst Watch/STV Downside call corresponds to a sell recommendation. Any stock not assigned to a Catalyst Watch Upside, Catalyst Watch Downside, STV Upside, or STV Downside call is considered Catalyst Watch/STV No View. For purposes of FINRA ratings distribution-disclosure rules, we correspond Catalyst Watch/STV No View to Hold in our ratings distribution table for our Catalyst Watch/STV Upside/Downside rating system. However, we reiterate that we do not consider No View to be a recommendation. For all Catalyst Watch/STV Upside/Downside calls, risk exists that the catalyst(s) and associated share-price movement will not materialize as expected.

RESEARCH ANALYST AFFILIATIONS / NON-US RESEARCH ANALYST DISCLOSURES
The legal entities employing the authors of this report are listed below (and their regulators are listed further herein). Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization (but are employed by an affiliate of the member organization) and therefore may not be subject to the FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Citigroup Global Markets Taiwan Securities Company Limited	Jack Chen; Michael Hung; Laura (Chia Yi) Chen; Nicholas Lai

OTHER DISCLOSURES

Any price(s) of instruments mentioned in recommendations are as of the prior day's market close on the primary market for the instrument, unless otherwise stated.

The completion and first dissemination of any recommendations made within this research report are as of the Eastern date-time displayed at the top of the Product. If the Product references views of other analysts then please refer to the price chart or rating history table for the date/time of completion and first dissemination with respect to that view.

Regulations in various jurisdictions require that where a recommendation differs from any of the author's previous recommendations concerning the same financial instrument or issuer that has been published during the preceding 12-month period that the change(s) and the date of that previous recommendation are indicated. For fundamental coverage please refer to the price chart or rating change history within this disclosure appendix or the issuer disclosure summary at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

Citi Research has implemented policies for identifying, considering and managing potential conflicts of interest arising as a result of publication or distribution of investment research. A description of these policies can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

The proportion of all Citi Research research recommendations that were the equivalent to "Buy","Hold","Sell" at the end of each quarter over the prior 12 months (with the % of these that had received investment firm services from Citi in the prior 12 months shown in brackets) is as follows: Q4 2025 Buy 33% (63%), Hold 44% (50%), Sell 23% (46%), RV 0.4% (91%); Q3 2025 Buy 33% (61%), Hold 44% (52%), Sell 23% (50%), RV 0.4% (80%); Q2 2025 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.4% (86%); Q1 2025 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.5% (87%). For the purposes of disclosing recommendations other than for equity (whose definitions can be found in the corresponding disclosure sections), "Buy" means a positive directional trade idea; "Sell" means a negative directional trade idea; and "Relative Value" means any trade idea which does not have a clear direction to the investment strategy.

European regulations require a 5 year price history when past performance of a security is referenced. CitiVelocity's Charting Tool (https://www.citivelocity.com/cv2/#go/CHARTING_3_Equities) provides the facility to create customisable price charts including a five year option. This tool can be found in the Data & Analytics section under any of the asset class menus in CitiVelocity (https://www.citivelocity.com/). For further information contact CitiVelocity support (https://www.citivelocity.com/cv2/go/CLIENT_SUPPORT). The source for all referenced prices, unless otherwise stated, is DataCentral, which sources price information from LSEG Data & Analytics. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance.

Investors should always consider the investment objectives, risks, and charges and expenses of an ETF carefully before investing. The applicable prospectus and key investor information document (as applicable) for an ETF should contain this and other information about such ETF. It is important to read carefully any such prospectus before investing. Clients may obtain prospectuses and key investor information documents for ETFs from the applicable distributor or authorized participant, the exchange upon which an ETF is listed and/or from the applicable website of the applicable ETF issuer. The value of the investments and any accruing income may fall or rise. Any past performance, prediction or forecast is not indicative of future or likely performance. Any information on ETFs contained herein is provided strictly for illustrative purposes and should not be deemed an offer to sell or a solicitation of an offer to purchase units of any ETF either explicitly or implicitly. The opinions expressed are those of the authors and do not necessarily reflect the views of ETF issuers, any of their agents or their affiliates.

Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Please be advised that pursuant to Executive Order 13959 as amended (the "Order"), U.S. persons are prohibited from investing in securities of any company determined by the United States Government to be the subject of the Order. This research is not intended to be used or relied upon in any way that could result in a violation of the Order. Investors are encouraged to rely upon their own legal counsel for advice on compliance with the Order and other economic sanctions programs administered and enforced by the Office of Foreign Assets Control of the U.S. Treasury Department.

This communication is directed at persons who are "Eligible Clients" as such term is defined in the Israeli Regulation of Investment Advice, Investment Marketing and Investment Portfolio Management law, 1995 (the "Advisory Law"). Within Israel, this communication is not intended for retail clients and Citi will not make such products or transactions available to retail clients or to non-Eligible Clients. The presenter is not licensed as investment advisor or investment marketer by the Israeli Securities Authority ("ISA") and this communication does not constitute investment or marketing advice. The information contained herein may relate to matters that are not regulated by the ISA. Any securities which are the subject of this communication may not be offered or sold to any Israeli person except pursuant to a security offering exemption according to the Israeli Securities Law, 1968and the public offering rules provided thereunder.

Citi Research broadly and simultaneously disseminates its research content to the Firm's institutional and retail clients via the Firm's proprietary electronic distribution platforms (e.g., Citi Velocity and various Global Wealth platforms). As a convenience, certain, but not all, research content may be distributed through third party aggregators. Clients may receive published research reports by email, on a discretionary basis, and only after such research content has been broadly disseminated. Certain research is made available only to institutional investors to satisfy regulatory requirements. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client's individual preferences as to the frequency and manner of receiving communications from analysts, the client's risk profile and investment focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with the Firm and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Resolução 20 and ASIC Regulatory Guide 264, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. Statements and views concerning ESG (environmental, social, governance) factors are typically based upon public statements made by the affected company or other public news, which the author may not have independently verified. ESG factors are one consideration that investors may choose to examine when making investment decisions. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental equity or credit research report, it is the intention of Citi Research to provide research coverage of the covered issuers, including in response to news affecting the issuer. For non-fundamental research reports, Citi Research may not provide regular updates to the views, recommendations and facts included in the reports. Notwithstanding that Citi Research maintains coverage on, makes recommendations concerning or discusses issuers, Citi Research may be periodically restricted from referencing certain issuers due to legal or policy reasons. Where a component of a published trade idea is subject to a restriction, the trade idea will be removed from any list of open trade ideas included in the Product. Upon the lifting of the restriction, the trade idea will either be re-instated in the open trade ideas list if the analyst continues to support it or it will be officially closed. Citi Research may provide different research products and services to different classes of customers (for example, based upon long-term or short-term investment horizons) that may lead to differing conclusions or recommendations that could impact the price of a security contrary to the recommendations in the alternative research product, provided that each is consistent with the rating system for each respective product.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by.

The Product is made available in **Australia** through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. Citigroup Global Markets Australia Pty Limited is not an Authorised Deposit-Taking Institution under the Banking Act 1959, nor is it regulated by the Australian Prudential Regulation Authority.

The Product is made available in **Brazil** by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários ("CVM"), BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBIMA – Associação Brasileira das Entidades dos Mercados Financeiro e de Capitais. Av. Paulista, 1111 - 14º andar(parte) - CEP: 01311920 - São Paulo - SP.

This Product is available in **Chile** through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Comisión Para El Mercado Financiero. Enrique Foster Sur, 20, piso 6, Las Condes, Santiago, Chile.

Disclosure for investors in the Republic of Colombia :This communication or message does not constitute a professional recommendation to make investment in the terms of article 2.40.1.1.2 of Decree 2555 de 2010 or the regulations that modify, substitute or complement it. Para la elaboración y distribución de informes de investigación y de comunicaciones generales de que trata este artículo no se requiere ser una entidad vigilada por la Superintendencia Financiera de Colombia.

The Product is made available in **Germany** by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt fur Finanzdienstleistungsaufsicht BaFin). Börsenplatz 9, 60313 Frankfurt am Main, Germany.

Unless otherwise specified, if the analyst who prepared this report is based in Hong Kong and it relates to “securities” (as defined in the Securities and Futures Ordinance (Cap.571 of the Laws of Hong Kong)), the report is issued in Hong Kong by Citigroup Global Markets Asia Limited. Citigroup Global Markets Asia Limited is regulated by Hong Kong Securities and Futures Commission. If the report is prepared by a non-Hong Kong based analyst, please note that such analyst (and the legal entity that the analyst is employed by or accredited to) is not licensed/registered in Hong Kong and they do not hold themselves out as such. Please refer to the section “Research Analyst Affiliations / Non-US Research Analyst Disclosures” for the details of the employment entity of the analysts.

The Product is made available in **India** by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking (SEBI Registration No. INM000010718) and stock brokerage ((SEBI Registration No. INZ000263033) in India, and is registered with SEBI in this regard. Registration granted by SEBI and certification from National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. CGM’s registered office is at 1202, 12th Floor, First International Financial Centre (FIFC), G Block, Bandra Kurla Complex, Bandra East, Mumbai – 400098 & registered Tel: +91 22 61759999. Citi maintains robust policies, procedures,controls, and training to ensure continued compliance with all applicable rulesand regulations. All recommendations contained herein are made by dulyqualified research analysts. CGM’s Corporate Identity Number is U99999MH2000PTC126657, and its Compliance Officer [Vishal Bohra] contact details are: Tel:+91-022-61759994, Fax:+91-

022-61759851, Email: cgmcompliance@citi.com. The Investor Charter in respect of Research Analysts, the Compliance Audit Report and Complaints information can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. The grievance officer [Niraj Mody] contact details are Tel: +91-22-42775002, Email: EMEA.CR.Complaints@citi.com. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. SEBI prescribed Client Terms & Conditions can be found at https://www.citivelocity.com/rendition/authfilelinksvcs/eppublic/V1/file?paramData=ZmlsZU5hbWU9L1MzL0dETVMvcHViRGlzY2xvc3VyZUh0bWxzL2dkbV9kaXNfc2ViaV9UZXJtc29mVXNlLmh0bWw

The Product is made available in **Indonesia** through PT Citigroup Sekuritas Indonesia. Citibank Tower 10/F, Pacific Century Place, SCBD lot 10, Jl. Jend Sudirman Kav 52-53, Jakarta 12190, Indonesia. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Services Authority (OJK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations.

The Product is made available in **Japan** by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Otemachi Park Building, 1-1-1 Otemachi, Chiyoda-ku, Tokyo 100-8132 Japan. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help.

The product is made available in the **Kingdom of Saudi Arabia** in accordance with Saudi laws through Citigroup Saudi Arabia, which is regulated by the Capital Market Authority (CMA) under CMA license (17184-31). 2239 Al Urubah Rd – Al Olaya Dist. Unit No. 18, Riyadh 12214 – 9597, Kingdom Of Saudi Arabia.

The Product is made available in **Korea** by Citigroup Global Markets Korea Securities Ltd. (CGMK), which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). The address of CGMK is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of CGMK. http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/DISFundMgrAnalystList.xml&divisionId=MDIS03002002000000&serviceId=SDIS03002002000. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. This research report is intended to be provided only to Professional Investors as defined in the Financial Investment Services and Capital Market Act and its Enforcement Decree in Korea.

The Product is made available in **Malaysia** by Citigroup Global Markets Malaysia Sdn Bhd (Registration No. 199801004692 (460819-D)) ("CGMM") to its clients and CGMM takes responsibility for its contents as regards CGMM's clients. CGMM is regulated by the Securities Commission Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product.

The Product is made available in **Mexico** by Citi México Casa de Bolsa, S.A. de C.V., Grupo Financiero Citi México which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comision Nacional Bancaria y de Valores. Prolongación Reforma 1196, 24 floor, Colonia Santa Fe, Alcaldía Cuajimalpa de Morelos, C.P. 05348, Ciudad de México.

The Product is made available in **Poland** by Biuro Maklerskie Banku Handlowego (DMBH), separate department of Bank Handlowy w Warszawie S.A. a subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Biuro Maklerskie Banku Handlowego (DMBH), ul.Senatorska 16, 00-923 Warszawa.

The Product is made available in **Singapore** through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services and Exempt Financial Advisor license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citi Private Bank, the Product is made available in Singapore by Citi Private Bank through Citibank, N.A., Singapore Branch. Citibank N.A., Singapore Branch is a licensed bank in Singapore that is regulated by the Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citibank Singapore Limited ("CSL"), the Product is distributed in Singapore by CSL to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of the Product. Please contact your Citigold//Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289).

Citigroup Global Markets (Pty) Ltd. is incorporated in the **Republic of South Africa** (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa.

The Product is made available in the **Republic of China (Taiwan)** through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14F, 15F and 16F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the license scope and the applicable laws and regulations in the Republic of China (Taiwan). CGMTS is regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China (Taiwan). No portion of the Product may be reproduced or quoted in the Republic of China (Taiwan) by the press or any third parties [without the written authorization of CGMTS]. Pursuant to the applicable laws and regulations in the Republic of China (Taiwan), the recipient of the Product shall not take advantage of such Product to involve in any matters in which the recipient may have conflicts of interest. If the Product covers securities which are not allowed to be offered or traded in the Republic of China (Taiwan), neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China (Taiwan). The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus.

The Product is made available in **Thailand** through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana ,Bangkok 10110, Thailand.

Disclosure for investors in the Republic of Turkey: The Product is made available in **Turkey** through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey. Under Capital Markets Law of Turkey (Law No: 6362), the investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Comments and recommendations stated here rely on the individual opinions of the ones providing these comments and recommendations. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations. Furthermore, Citi Research is a division of Citigroup Global Markets Inc. (the “Firm”), which does and seeks to do business with companies and/or trades on securities covered in this research reports. As a result, investors should be aware that the Firm may have a conflict of interest that could affect the objectivity of this report, however investors should also note that the Firm has in place organisational and administrative arrangements to manage potential conflicts of interest of this nature.

In the **U.A.E**, these materials (the "Materials") are communicated byCitigroup Global Markets Limited, DIFC branch ("CGML"), an entityregistered in the Dubai International Financial Center ("DIFC") andlicensed and regulated by the Dubai Financial Services Authority ("DFSA",license #CL0221) to Professional Clients and Market Counterparties, as definedin DFSA regulations, only and should not be relied upon or distributed toRetail Clients. Financial products and/or services to which the Materialsrelate will only be made available to Professional Clients and MarketCounterparties. Citigroup Global Markets Limited DIFC Branch registered addressis Level 3, Gate District Building 02, Dubai International Financial Centre andcan be contacted on +971 4 509 97 90.

The Product is made available in **United Kingdom** by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority (“PRA”) and regulated by the Financial Conduct Authority (“FCA”) and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB.

The Product is made available in **United States** and **Canada** by Citigroup Global Markets Inc., which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013.

Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht-BaFin).

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.

Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested.The yield and average life of CMOs (collateralized mortgage obligations) referenced in this Product will fluctuate depending on the actual rate at which mortgage holders prepay the mortgages underlying the CMO and changes in current interest rates. Any government agency backing of the CMO applies only to the face value of the CMO and not to any premium paid. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt, investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Card Insights. Where this report references Card Insights data, Card Insights consists of selected data from a subset of Citi’s proprietary credit card transactions. Such data has undergone rigorous security protocols to keep all customer information confidential and secure; the data is highly aggregated and anonymized so that all unique customer identifiable information is removed from the data prior to receipt by the report's author or distribution to external parties. This data should be considered in the context of other economic indicators and publicly available information. Further, the selected data represents only a subset of Citi's proprietary credit card transactions due to the selection methodology or other limitations and should not be considered as indicative or predictive of the past or future financial performance of Citi or its credit card business.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes the Firm's estimates, data from company reports and feeds from LSEG Data & Analytics. The source for all referenced prices, unless otherwise stated, is DataCentral. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance. The printed and printable version of the research report may not include all the information <(e.g. certain financial summary information and comparable company data) that is linked to the online version available on the Firm's proprietary electronic distribution platforms.

Where included in this report, MSCI sourced information is the exclusive property of Morgan Stanley Capital International Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced, redisseminated or used to create any financial products, including any indices. This information is provided on an "as is" basis. The user assumes the entire risk of any use made of this information. MSCI, its affiliates and any third party involved in, or related to, computing or compiling the information hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the information have any liability for any damages of any kind. MSCI, Morgan Stanley Capital International and the MSCI indexes are services marks of MSCI and its affiliates. Where data is attributed to Morningstar that data is © 2026 Morningstar, Inc. All Rights Reserved. That information: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information.

The Firm accepts no liability whatsoever for the actions of third parties. The Product may provide the addresses of, or contain hyperlinks to, websites. Except to the extent to which the Product refers to website material of the Firm, the Firm has not reviewed the linked site. Equally, except to the extent to which the Product refers to website material of the Firm, the Firm takes no responsibility for, and makes no representations or warranties whatsoever as to, the data and information contained therein. Such address or hyperlink (including addresses or hyperlinks to website material of the Firm) is provided solely for your convenience and information and the content of the linked site does not in any way form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

© 2026 Citigroup Global Markets Inc. Citi Research is a division of Citigroup Global Markets Inc. Citi and Citi and Arc Design are trademarks and service marks of Citigroup Inc. and its affiliates and are used and registered throughout the world. All rights reserved. The research data in this report are not intended to be used for the purpose of (a) determining the price of or amounts due in respect of (or to value) one or more financial products or instruments and/or (b) measuring or comparing the performance of, or defining the asset allocation of a financial product, a portfolio of financial instruments, or a collective investment undertaking, and any such use is strictly prohibited without the prior written consent of Citi Research. Any unauthorized use, duplication, redistribution or disclosure of this report (the "Product"), including, but not limited to, redistribution of the Product by electronic mail, posting of the Product on a website or page, and/or providing to a third party a link to the Product, is prohibited by law and will result in prosecution. The information contained in the Product is intended solely for the recipient and may not be further distributed by the recipient to any third party.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST